

Solution Manual

Analysis and Design of Elementary MOS Amplifier Stages

**- First Edition -
Chapter 1**

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restart : with(Units[Standard]) : with(plots) : interface(imaginaryunit=j) : read "Parameters.mpl" :

1.1

(a) Note that this is a unilateral circuit. Hence, there is no need to worry about input/output port termination when computing R_{in} and R_{out} . By inspection:

$$R_{in} = r_1 + r_i :$$

$$R_{out} = r_2 + r_o :$$

(b) To find A_v , we apply an ideal test voltage at the input and measure the open circuit output voltage.

$$v_i = v_t \cdot \frac{r_i}{r_i + r_1} :$$

$$v_{oc} = -g_m \cdot v_i \cdot r_o = -g_m \cdot \frac{r_i \cdot r_o}{r_i + r_1} \cdot v_t :$$

$$A_v = \frac{v_{oc}}{v_t} = -g_m \cdot \frac{r_i \cdot r_o}{r_i + r_1} :$$

The voltage amplifier model is the circuit shown in Figure 1-9(a) with A_v , R_{in} , R_{out} as computed above.

(c) To find A_i , we apply an ideal test current at the input and measure the short circuit output current.

$$v_i = i_t \cdot r_i :$$

$$i_{sc} = g_m \cdot v_i \cdot \frac{r_o}{r_o + r_2} = g_m \cdot \frac{r_o \cdot r_i}{r_o + r_2} \cdot i_t :$$

$$A_i = \frac{i_{sc}}{i_t} = g_m \cdot \frac{r_o \cdot r_i}{r_o + r_2} :$$

The voltage amplifier model is the circuit shown in Figure 1-9(b) with A_i , R_{in} , R_{out} as computed above.

(d) To find G_m , we apply an ideal test voltage at the input and measure the short circuit output current.

$$v_i = v_t \cdot \frac{r_i}{r_i + r_1} :$$

$$i_{sc} = g_m \cdot v_i \cdot \frac{r_o}{r_o + r_2} = g_m \cdot \frac{r_o}{r_o + r_2} \cdot \frac{r_i}{r_i + r_1} \cdot v_t :$$

$$G_m = \frac{i_{sc}}{v_t} = g_m \cdot \frac{r_o}{r_o + r_2} \cdot \frac{r_i}{r_i + r_1} :$$

The transconductance amplifier model is the circuit shown in Figure 1-9(c) with G_m , R_{in} , R_{out} as computed above.

(e) To find R_m , we apply an ideal test current at the input and measure the open circuit output voltage.

$$v_i = i_t \cdot r_i :$$

$$v_{oc} = -g_m \cdot v_i \cdot r_o = -g_m \cdot r_i \cdot r_o \cdot i_t :$$

$$R_m = \frac{v_{oc}}{i_t} = -g_m \cdot r_i \cdot r_o :$$

The transresistance amplifier model is the circuit shown in Figure 1-9(d) with R_m , R_{in} , R_{out} as computed above.

1.2

For the circuit in Figure 1-10 we have:

$$v_{in} = v_s \cdot \frac{R_{in}}{R_{in} + R_S} :$$

$$v_{out} = A_v \cdot v_{in} \cdot \frac{R_L}{R_L + R_{out}} :$$

$$A'_v = \frac{v_{out}}{v_s} = A_v \cdot \frac{R_{in}}{R_{in} + R_S} \cdot \frac{R_L}{R_L + R_{out}} :$$

For the circuit in Figure 1-11 we find:

$$v_{in} = v_s \cdot \frac{R_{in}}{R_{in} + R_S} :$$

$$v_{out} = -G_m \cdot \frac{R_{out} \cdot R_L}{R_{out} + R_L} = \frac{A_v}{R_{out}} \cdot \frac{R_{out} \cdot R_L}{R_{out} + R_L}$$

$$A'_v = \frac{v_{out}}{v_s} = A_v \cdot \frac{R_{in}}{R_{in} + R_S} \cdot \frac{R_L}{R_L + R_{out}} :$$

1.3

(a) The given circuit corresponds to the model of Figure 1-9(a) with the following parameters:

$$R_{in} = r_1 + r_i :$$

$$R_{out} = r_2 + r_o :$$

$$A_v = -g_m \cdot \frac{r_i \cdot r_o}{r_i + r_1} :$$

The overall voltage gain is:

$$A'_v = \frac{v_{out}}{v_s} = A_v \cdot \frac{R_{in}}{R_{in} + R_S} \cdot \frac{R_L}{R_L + R_{out}} = -g_m \cdot \frac{r_i \cdot r_o}{r_i + r_1} \cdot \frac{r_1 + r_i}{r_1 + r_i + R_S} \cdot \frac{R_L}{R_L + r_2 + r_o}$$
$$A'_v = -g_m \cdot \frac{r_i \cdot r_o}{r_1 + r_i + R_S} \cdot \frac{R_L}{R_L + r_2 + r_o}$$

(b) To increase the overall voltage gain:

We need to decrease r_1 , since it appears in the denominator. We can also argue intuitively from the circuit in Figure P1-1, that this must be true since smaller r_1 increases v_i .

We need to decrease r_2 , since it appears in the denominator. We can also argue intuitively from the circuit in Figure P1-1, that this must be true since smaller r_2 reduces the voltage division with R_L .

We need to increase r_i and r_o . These terms appear both in the numerator and denominator, but the denominator is a sum that grows more slowly than the numerator. This can be shown formally by taking the partial derivatives. Intuitively, we see that larger r_i will help increase v_i and larger r_o means that less current "is stolen" from the load.

1.4

(a) The given circuit corresponds to the model of Figure 1-9(b) with the following parameters:

$$R_{in} = r_1 + r_i :$$

$$R_{out} = r_2 + r_o :$$

$$A_i = g_m \cdot \frac{r_o \cdot r_i}{r_o + r_2} :$$

The overall current gain is:

$$A'_i = \frac{i_{out}}{i_s} = A_i \cdot \frac{R_S}{R_{in} + R_S} \cdot \frac{R_{out}}{R_L + R_{out}} = g_m \cdot \frac{r_o \cdot r_i}{r_o + r_2} \cdot \frac{R_S}{r_1 + r_i + R_S} \cdot \frac{r_2 + r_o}{R_L + r_2 + r_o}$$
$$A'_i = g_m \cdot \frac{r_o \cdot r_i \cdot R_S}{r_1 + r_i + R_S} \cdot \frac{1}{R_L + r_2 + r_o}$$

(b) To increase the overall current gain:

We need to decrease r_1 , since it appears in the denominator. We can also argue intuitively from the circuit in Figure P1-1, that this must be true since smaller r_1 increases v_i .

We need to decrease r_2 , since it appears in the denominator. We can also argue intuitively from the circuit in Figure P1-1, that this must be true since smaller r_2 improves the current division between r_o and R_L .

We need to increase r_i and r_o . These terms appear both in the numerator and denominator, but the denominator is a sum that grows more slowly than the numerator. This can be shown formally by taking the partial derivatives. Intuitively, we see that larger r_i will help increase v_i and larger r_o means that less current "is stolen" from the load.

1.5

Note: The problem is unnecessarily complex with r_i , r_o , r_1 and r_2 included. The solution below is worked with r_i , r_o removed and $r_1=r_2=0$. This was also noted in the errata.

(a) Since this circuit is bilateral, we must add R_L in the calculation of the input resistance. We apply a test voltage at the input port and measure the test current flowing out of this source.

$$Eq1 := 0 = \frac{v_t - v_{out}}{r_f} - i_t :$$

$$Eq2 := 0 = g_m \cdot v_t + \frac{v_{out} - v_t}{r_f} + \frac{v_{out}}{R_L} :$$

$$\text{simplify}(\text{solve}(\{Eq1, Eq2\}, [v_p, v_{out}])) = \left[\left[v_t = \frac{i_t (R_L + r_f)}{g_m R_L + 1}, v_{out} = -\frac{R_L i_t (g_m r_f - 1)}{g_m R_L + 1} \right] \right]$$

The input resistance is therefore:

$$R_{in} = \frac{v_t}{i_t} = \frac{r_f + R_L}{1 + g_m R_L} :$$

To measure the output resistance, we connect R_S across the input port and apply a test source at the output port.

$$Eq1 := 0 = \frac{v_i}{R_S} + \frac{v_i - v_t}{r_f} :$$

$$Eq2 := 0 = g_m \cdot v_i + \frac{v_t - v_i}{r_f} - i_t :$$

$$\text{simplify}(\text{solve}(\{Eq1, Eq2\}, [v_p, v_i])) = \left[\left[v_t = \frac{(r_f + R_S) i_t}{R_S g_m + 1}, v_i = \frac{R_S i_t}{R_S g_m + 1} \right] \right]$$

The output resistance is therefore:

$$R_{out} = \frac{v_t}{i_t} = \frac{r_f + R_S}{1 + g_m R_S} :$$

(b) To find A_v , we apply an ideal test voltage at the input and measure the open circuit output voltage.

$$v_{oc} = v_t - g_m v_t \cdot r_f :$$

$$A_v = \frac{v_{oc}}{v_t} = 1 - g_m r_f :$$

The voltage amplifier model is the circuit shown in Figure 1-9(a) with A_v , R_{in} , R_{out} as computed above.

(c) To find A_i , we apply an ideal test current at the input and measure the short circuit output current.

$$v_i = i_t \cdot r_f :$$

$$i_{sc} = g_m v_i - i_t :$$

$$A_i = \frac{i_{sc}}{i_t} = g_m r_f - 1 :$$

The current amplifier model is the circuit shown in Figure 1-9(b) with A_i , R_{in} , R_{out} as computed above.

(d) To find G_m , we apply an ideal test voltage at the input and measure the short circuit output current.

$$i_{sc} = g_m v_t - \frac{v_t}{r_f} :$$

$$G_m = \frac{i_{sc}}{v_t} = g_m - \frac{1}{r_f} :$$

The transconductance amplifier model is the circuit shown in Figure 1-9(c) with G_m , R_{in} , R_{out} as computed above.

(e) To find R_m , we apply an ideal test current at the input and measure the open circuit output voltage.

$$g_m v_i = i_t :$$

$$v_{oc} = v_i - i_t r_f :$$

$$R_m = \frac{v_{oc}}{i_t} = -r_f + \frac{1}{g_m} :$$

The transresistance amplifier model is the circuit shown in Figure 1-9(d) with R_m , R_{in} , R_{out} as computed above.

1.6

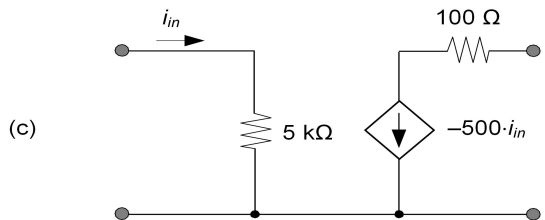
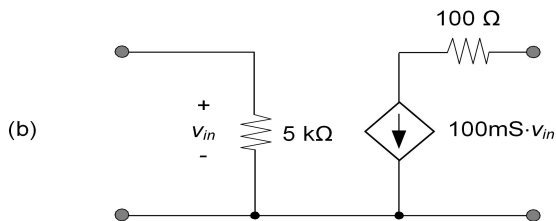
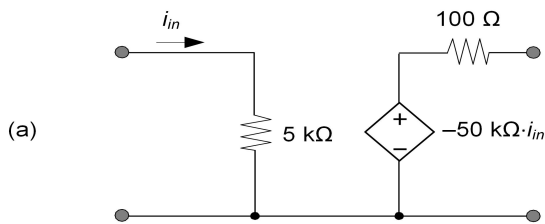
$$A_v := 10 : R_{in} := 5 \text{ [k}\Omega\text{]} : R_{out} := 100 \text{ [}\Omega\text{]} :$$

R_{in} and R_{out} are the same in all models. The gain elements for the alternate representations are found as follows.

$$R_m = \frac{v_{oc}}{i_{in}} = \frac{A_v \cdot v_{in}}{\frac{v_{in}}{R_{in}}} = -A_v \cdot R_{in} : \quad R_m := -A_v \cdot R_{in} = -50 \text{ [k}\Omega\text{]}$$

$$G_m = \frac{i_{sc}}{v_{in}} = \frac{-\frac{A_v \cdot v_{in}}{R_{out}}}{v_{in}} = -\frac{A_v}{R_{out}} : \quad G_m := -\frac{A_v}{R_{out}} = -\frac{1}{10} \text{ [S]} \xrightarrow{\text{replace units}} -100 \text{ [mS]}$$

$$A_i = \frac{i_{sc}}{i_{in}} = \frac{-\frac{A_v \cdot v_{in}}{R_{out}}}{\frac{v_{in}}{R_{in}}} = -A_v \cdot \frac{R_{in}}{R_{out}} : \quad A_i := -A_v \cdot \frac{R_{in}}{R_{out}} = -500$$



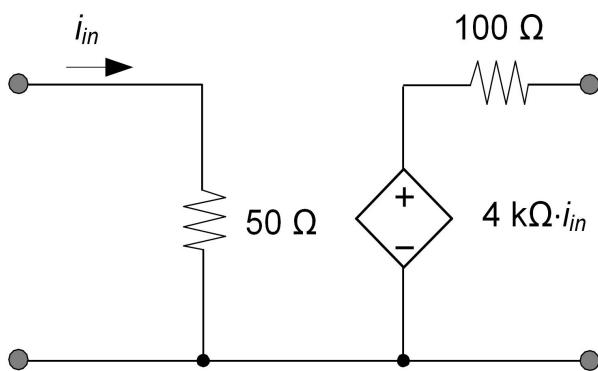
1.7

$$A_{i1} := 1 : G_{m2} := 10 \text{ [mS]} : A_{v3} := 0.8 : R_{in1} := 50 \text{ [}\Omega\text{]} : R_{out1} := 500 \text{ [}\Omega\text{]} : R_{out2} := 1 \text{ [k}\Omega\text{]} : R_{out3} := 100 \text{ [}\Omega\text{]} :$$

R_{in} and R_{out} are simply equal to R_{in1} and R_{out3} , respectively. The transresistance of the circuit is computed as follows.

$$R_m = \frac{v_{oc}}{i_{in}} = \frac{A_{v3} \cdot v_{in3}}{i_{in}} = \frac{A_{v3} \cdot (-G_{m2} \cdot v_{in2} \cdot R_{out2})}{i_{in}} = \frac{A_{v3} \cdot (-G_{m2} \cdot R_{out2}) \cdot (-A_{i1} \cdot i_{in} \cdot R_{out1})}{i_{in}} :$$

$$R_m := A_{v3} \cdot G_{m2} \cdot R_{out2} \cdot A_{i1} \cdot R_{out1} = 4000.000 \text{ [}\Omega\text{]}$$



1.8

$$R_1 := 1 \llbracket k\Omega \rrbracket : g_m := 1 \llbracket mS \rrbracket : R_2 := 100 \llbracket k\Omega \rrbracket : R_S := 10 \llbracket k\Omega \rrbracket : R_L := 10 \llbracket k\Omega \rrbracket :$$

Note: The values for RS and RL were missing in the problem statement of the first edition (as noted in the errata file).

Aif, Rin and Rout follow from the expressions derived in Example 1-3:

$$A'_{if} := -1 :$$

$$R_{in} := \frac{1 + \frac{R_L}{R_2}}{g_m + \frac{1}{R_2}} = 1089.109 \llbracket \Omega \rrbracket$$

$$R_{out} := R_2 + R_S + g_m \cdot R_2 \cdot R_S = 1110 \llbracket k\Omega \rrbracket$$

We find A_{ir} by injecting a test current into the output port and measuring the short circuit current at the input port. Due to the short at the input, $v_{in}=0$ and the gm generator has zero current. Hence, the injected current flows directly from the output port back out of the input port, i.e.

$$A'_{ir} := -1 :$$

Thus, A_{if} and A_{ir} are identical. However, it is important to note that the circuit is a much better current amplifier in the forward direction than in the reverse direction, due to the large difference in R_{in} and R_{out} . One can show this by comparing the overall current gain from the input to the output and from the output back to the input.